

Jiasheng (Jason) Hu

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I am a 4th year PhD student in the department of Computer Science at University of Toronto, member of [Far Data Lab](#), supervised by Prof. [Qizhen Zhang](#). My research interests mainly lie in cloud-native and distributed data processing systems, particularly their intersection with emerging hardware such as DPUs (SmartNICs). The amount of data to be processed keeps growing, yet the CPU compute power is not keeping up – in short, I enjoy building and optimizing distributed data processing systems, with a focus on the network communication aspect.

EDUCATION

PhD Student in Computer Science, University of Toronto

September 2023 – Present

4th Year PhD Student @ Far Data Lab, supervised by Prof. Qizhen Zhang.

BEng in Computer Science, First Class Honours, University of Edinburgh

September 2019 – May 2023

Honours Project: Homa transport protocol with zero-copy networking under Linux

RESEARCH & PUBLICATIONS

Making Sense of DPU Performance for Cloud Data Processing

SoCC 2026

Jiasheng Hu, Chihan Cui, Yuanfan Cehn, Philip A. Bernstein, Jialin Li, Qizhen Zhang

We present our benchmark results of recent mass-production DPUs to understand their performance characteristics for data processing tasks, and in particular, detailed investigations into DPU offloading potential for database workloads.

dpKernels: Harvesting DPU Compute Resources for Data-path Efficiency in Cloud Data Processing

VLDB 2026

Jiasheng Hu, Kaiwen Zheng, Anna Li, Sidharth Sankhe, Philip A. Bernstein, Qizhen Zhang

We introduce two levels of abstraction: dpKernels and dpManager, for achieving unified, efficient, and portable primitives that abstract DPU compute resources (i.e., hardware accelerators and SoC cores) for cloud data systems to deliver their promises with optimized, scheduled, and cross-platform executions.

Tuning the Lookahead Distance for PostgreSQL Asynchronous IO

VLDB 2026 Industrial Track

Wentao Wu, Jiasheng Hu, Manoj Syamala, Andres Freund, Vivek Narasayya

The storage subsystem adopts a classic producer/consumer model, but poses a critical performance challenge: every read stream needs to decide the “lookahead distance”. We address this challenge by proposing an adaptive approach to automatically adjust the lookahead distance by only leveraging feedback information based on the IO completion time, inspired by and draws a connection to the theory of optimal flow control in computer networking.

PD3: Prefetching Data with DPUs for Disaggregated Memory

NSDI 2026

Sidharth Sankhe, Felix Zhang, Umayrah Chonee, Sherman Lim, Jiasheng Hu, Jialin Li, Qizhen Zhang

We introduce PD3, a memory disaggregation solution that avoids cache misses, via prefetching, on compute servers and thus all their associated overhead, using DPUs, which prevents mispredictions with network support and minimal yet critical application information.

DPDPU: Data Processing with DPUs ([paper](#))

CIDR 2025

Jiasheng Hu, Philip A. Bernstein, Jialin Li, Qizhen Zhang

A vision paper where opportunities of utilizing DPUs to tackle existing problems in today’s data processing systems are discussed, proposing the DPDPU framework, and addressing the related challenges and obstacles.

DDS: DPU-optimized Disaggregated Storage ([paper](#))

VLDB 2024

Qizhen Zhang, Philip Bernstein, Badrish Chandramouli, Jason Hu, Yiming Zheng

One of the pioneering work in using DPU to accelerate Database systems, overhead and optimization opportunities within existing systems are identified, and we proposed partial offloading of read operations using DPU and the DDS framework, which resulted in an order of magnitude latency reduction, as well as up to tens of host CPU cores savings.

Combining Homa protocol with zero-copy networking ([code](#)) September 2022 - March 2023
Honours project, supervised by Prof. Michio Honda @ University of Edinburgh. Incorporating the Homa transport protocol in Linux Kernel, such as NVMe-oF and sendpage API; evaluating performance benefits of Homa for applications, such as gRPC and Redis.

ACADEMIC SERVICES

Reviewer, IEEE Transactions on Computers (2025)

Reviewer, IEEE Transactions on Parallel and Distributed Systems (2025)

TALKS AND PRESENTATIONS

DPDPU: Data Processing with DPUs

- **VLDB 2024**, Guangzhou, China; **OnDB 2024**, University of Waterloo, Canada.

dpKernels: Harvesting DPU Compute Resources for Data-path Efficiency in Cloud Data Processing

- **VLDB 2026**, Boston, MA, USA.

INDUSTRIAL RESEARCH & WORK EXPERIENCE

Research Intern, Microsoft Research, Redmond, WA (Summer 2025)

- **Responsibilities:** Conducting research and related projects at the Data Systems Group at Microsoft Research, under the supervision of [Wentao Wu](#) as a research intern.
- **Project:** Tuning the Lookahead Distance for PostgreSQL Asynchronous IO (paper accepted to VLDB 2026): optimizing the storage engine performance by adaptively choosing an optimal lookahead distance for prefetching database pages; paper submitted to VLDB 2026 industry track.

TEACHING EXPERIENCES

Teaching Assistant at University of Toronto

May 2024 - present

- Introduction to Databases (CSC343, CSCC43), Introduction to Computer Programming (CSC108), Mathematical Expression and Reasoning for Computer Science (CSC165)

Teaching Assistant and Volunteer Mentor at University of Edinburgh

September 2021 - April 2023

- Introduction to Computation, System Design Project